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Benzoyl Peroxide-Based Combination Therapies for Acne Vulgaris

A Comparative Review

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Contents

Abstract

Benzoyl peroxide, with its broad-spectrum antimicrobial activity, is among the most widely used topical agents in the treatment of inflammatory acne vulgaris. Benzoyl peroxide is marketed either alone or in combination with other topical antibiotics; namely, erythromycin and clindamycin. The combination products confer specific advantages over benzoyl peroxide alone, particularly in decreasing the *in vivo* follicular counts of *Propionibacterium acnes*, the anaerobic bacterium implicated in the pathogenesis of acne. In addition, the topical treatment of inflammatory acne has been complicated by the development of *P acnes* resistance to topical erythromycin and clindamycin. Combination products containing benzoyl peroxide and the topical antibiotics have been shown to both: (i) prevent the development of antibiotic resistance in acne patients; and (ii) confer significant clinical improvement to patients who have already developed antibiotic resistance.

Acne vulgaris, a chronic inflammatory disease of the pilosebaceous follicle, is the most prevalent skin disorder and is estimated to affect 85–100% of the world's population at some time in their lives.[1] It most commonly arises during adolescence when there is a sharp rise in the production of androgens, but an increasing number of patients – women, in particular – complain of the appearance of lesions in the third to fifth decades of life. The pathogenesis of acne is now understood to be multifactorial, involving four main factors that explain the wide variation in lesion morphology and clinical severity: (i) an androgen-stimulated increase in sebum production; (ii) hyperproliferation and abnormal desquamation of the intrafollicular epithelium, leading to follicular hyperkeratinization and the precursor lesion (the microcomedo); (iii) proliferation of the anaerobic diphtheroid, *Propionibacterium acnes*, within the sebaceous follicle; and (iv) inflammation resulting from the release of proinflammatory cytokines by *P acnes* or from the rupture of the microcomedo, explaining the papules and pustules of inflammatory acne.[1]

Acne affects the areas of the body that are richest in sebaceous glands; namely, the face, upper back, chest and shoulders. Inflammation and physical manipulation of lesions can lead to post-inflammatory erythema, hyperpigmentation and permanent scarring, which can have profound psychologic consequences for the patient. Successful treatment aimed at reducing the risk of scarring is, therefore, of the utmost importance and is based on our current understanding of the pathophysiology of acne vulgaris.

1. Medications Used in the Treatment of Acne Vulgaris

A variety of over-the-counter and prescription-only treatment options are available for the treatment of acne vulgaris, including both topical and systemic therapies. Topical agents that provide comedolytic activity utilize retinoids and, to a lesser extent, salicylic acid. Topical retinoids are known to normalize the abnormal desquamation within the follicle as well as decrease inflammation and improve the appearance of post-inflammatory hyperpigmentation. Topical erythromycin and clindamycin are used mainly for their action as topical antibiotics in reducing the follicular population of *P acnes* and decreasing inflammatory lesion counts. Benzoyl peroxide is antibacterial and indirectly comedolytic when applied topically. Intralesional injections of corticosteroids rapidly decrease the inflammation associated with papules and pustules. Recently, phototherapy with blue (415nm) and red (660nm) light and photodynamic therapy with topical aminolevulinic acid have emerged as new options in the treatment of inflammatory acne.[2]

Systemic medications used in the treatment of acne include oral antibiotics with activity against *P acnes*. Oral antibiotics are also believed to have anti-inflammatory effects that are independent of their antibacterial properties. Oral contraceptives alone or in combination with spironolactone are used to modulate the hormonal influences on acne. In some cases, low-dose corticosteroids are used to treat resistant inflammatory disease. Isotretinoin is the only medication that is known to affect all four pathophysiologic factors responsible for acne and, in many cases, induces long-term remission.[1,3] Its adverse effect profile, however, precludes its use in all but the most severe or refractory acne.

2. Benzoyl Peroxide and Acne

The use of benzoyl peroxide was introduced in the 1960s and is, arguably, the most widely used topical agent for acne today, being available in both over-the-counter and prescription-only formulations in concentrations between 2.5% and 10%. In 1977, Kligman et al.[4] noted that the powerful broad-spectrum antimicrobial activity of benzoyl peroxide made it useful in the treatment of a variety of cutaneous diseases associated with bacterial or fungal infection and colonization, including acne. In the past, benzoyl peroxide has been used successfully in concentrations of up to 20% in the treatment of acne.[5]

Benzoyl peroxide is a lipophilic molecule that is able to penetrate the stratum corneum and enter the pilosebaceous follicle, where it is rapidly degraded to form benzoic acid.[6] In the follicle, before it is degraded, benzoyl peroxide is a powerful generator of free radicals that oxidize proteins in bacterial cell membranes. There is a rapid decrease in population counts of *P acnes* in the sebaceous follicle after the topical application of benzoyl peroxide; this is the basis for its activity in inflammatory acne. Benzoyl peroxide also has mild comedolytic properties, and its use leads to decreases in both inflammatory and non-inflammatory lesion counts. There are conflicting data on the ability of benzoyl peroxide to affect sebum production.[7,8] The general belief is that benzoyl peroxide does not suppress sebum production.[9]

3. Combination Benzoyl Peroxide Formulations

Benzoyl peroxide has played an important role in the topical treatment of acne for several decades, but its use is associated with a certain amount of allergic and irritant contact dermatitis, which limits patient compliance and thus decreases its potential efficacy. It has also been known to bleach hair and colored fabric. Over the years, a number of strategies have been employed in the hope of increasing efficacy and patient compliance without increasing irritation or other adverse effects.

In 1986, Mills et al.[10] reported results of three double-blind studies that showed that a 2.5% benzoyl peroxide formulation was as effective as the 5% and 10% formulations in reducing inflammatory lesions in acne vulgaris.

In the mid-to-late 1980s, there were several reports in the European literature of the use of a lotion combining 5% benzoyl peroxide and 2% miconazole nitrate in the treatment of acne. This combination was reported to be more effective at reducing inflammatory lesions than 5% benzoyl peroxide alone but the adverse effect profiles were found to be comparable.[11,12] This product is currently being used as a prescription topical medication indicated for mild-to-moderate inflammatory acne vulgaris, but is not marketed in the US.

In the US, the main combination products combine benzoyl peroxide with topical antibiotics. In 1991, a topical preparation consisting of benzoyl peroxide and meclocycline was reported to be less effective than benzoyl peroxide alone but more effective than topical meclocycline alone, in the treatment of inflammatory acne. However, the combination treatment was better tolerated than benzoyl peroxide alone.[13] Currently, the main topical antibiotics used in the treatment of acne in the US are erythromycin and clindamycin.

A recent review by Goulden[14] suggests the use of topical benzoyl peroxide in combination with oral antibiotics for the treatment of moderate acne vulgaris in adolescents as a way of decreasing the incidence of antibiotic resistance, which is an increasing problem with the use of oral antibiotics in acne management.[14]

In 2000, the possibility of biological synergism between mixtures of benzoyl peroxide and antibiotics was suggested when it was shown in an article by Burkhart et al.[15] that the production of free radicals by benzoyl peroxide was increased in the presence of macrolide antibiotics. The authors proposed that if increased free radical formation correlates with enhanced biological effect, their results might explain the increased efficacy of the combination products. The article also suggested that the tertiary amines contained in certain antibiotics may play a role in the catalysis of benzoyl peroxide free radical formation. This implies that benzoyl peroxide may also work synergistically with other tertiary amine-containing antibiotics to provide benefit in the treatment of acne.

3.1 Benzoyl Peroxide and Erythromycin

In early 1983, Burke et al.[16] published the results of a double-blind clinical trial comparing the efficacy of a 5% benzoyl peroxide gel with that of a 1.5% erythromycin lotion. They reported that the formulations were equally efficacious in reducing the number of small inflamed lesions and overall acne severity, but that the benzoyl peroxide gel also decreased the number of non-inflamed lesions while the erythromycin lotion had no effect on non-inflamed lesions.

Later in 1983, Chalker et al.[17] reported that a combination gel containing 3% erythromycin and 5% benzoyl peroxide was more effective than either product alone in reducing papules and pustules in a double-blind, controlled clinical trial. In addition, each topical agent alone was superior in efficacy to the vehicle control. The adverse effect profile of the combination gel was similar to that of benzoyl peroxide alone.

Recently, a randomized, double-blind, multicenter, parallel group study comparing the efficacies of a topical 3% erythromycin/5% benzoyl peroxide gel with a 0.025% tretinoin/4% erythromycin product showed that the benzoyl peroxide combination product compared very favorably in the treatment of moderate acne vulgaris of the face.[18] The authors reported significantly greater reductions in both physician- and patient-rated severity of acne symptoms with the benzoyl peroxide combination product compared with the tretinoin combination product as early as week 2 of treatment.

Antibiotic resistance is a well-documented consequence of systemic antibiotic use. In 1992, Harkaway et al.[19] noted a dramatic increase in the erythromycin-resistant populations of *Staphylococcus epidermidis* in the aerobic flora of skin treated with topical erythromycin for 12 weeks. The *S epidermidis* was found to be resistant not only to erythromycin, but also to clindamycin and tetracycline. However, skin treated with a combination of erythromycin and benzoyl peroxide, or benzoyl peroxide alone, resulted in a significant decrease in the number of aerobic bacteria without a concomitant change in the sensitivity of these organisms to erythromycin or other antibacterials. Later studies have also shown that resistant strains of *P acnes* develop after the use of topical erythromycin alone, whereas the combination of benzoyl peroxide and erythromycin counteracts the selection of erythromycin resistance and reduces the number of pre-existing resistant strains.[20] Benzoyl peroxide alone has never been associated with an increase in antibiotic resistance.

A combination product containing 3% erythromycin and 5% benzoyl peroxide is currently available as a prescription treatment indicated for mild-to-moderate acne. However, as benzoyl peroxide is such a powerful generator of reactive oxygen species, stability becomes an issue when it is used in combination with another molecule. The erythromycin/benzoyl peroxide combination product is unstable at high temperatures; it must be compounded at the time of dispensing and refrigerated to maintain its stability. In addition, its stability is not guaranteed after the expiration date and patients are advised to discontinue use 3 months after the product has been dispensed. To overcome these stability issues, a single-use erythromycin/benzoyl peroxide combination package has been developed that does not require compounding by the pharmacist. This new package has been shown to be well-tolerated and to have equivalent efficacy when compared with the original formulation.[21]

3.2 Benzoyl Peroxide and Clindamycin

In 1988, topical 1% clindamycin was shown, in a randomized, investigator-blind clinical trial, to be clinically effective in reducing acne lesion counts, but less so than 5% benzoyl peroxide gel. However, the clindamycin solution was much better tolerated than the benzoyl peroxide gel.[22] Since then, there have been several studies documenting the increased efficacy of a product combining 5% benzoyl peroxide with 1% clindamycin over 1% clindamycin alone in the reduction of *P acnes*, clindamycin-resistant *P acnes,* and total acne lesion counts. Again, the safety and adverse effect profiles of the combination products were comparable to those of 5% benzoyl peroxide gel alone.[23,24] However, there have also been isolated reports of diarrhea and pseudomembranous colitis associated with the use of topical clindamycin.[25,26]

A study comparing a combination product containing benzoyl peroxide and clindamycin with another combination formulation of benzoyl peroxide with erythromycin concluded that the two combination products were of equivalent efficacy and demonstrated similar tolerability.[27]

Unlike erythromycin/benzoyl peroxide combinations, the current combination therapies with benzoyl peroxide and clindamycin are stable at room temperature and, therefore, do not require refrigeration. However, some formulations still require reconstitu-

Table I. Comparison of available benzoyl peroxide (BP)-based combination therapies

Parameter	BP + erythromycin	BP + clindamycin
Percentage available in products	5% BP, 3% erythromycin	5% BP, 1% clindamycin
Comparative efficacy	Better than either product alone	Better than either product alone
Effect on development of antibacterial resistance	Decreases incidence over erythromycin alone; reduces counts of pre-existing resistant strains	Decreases incidence over clindamycin alone; reduces counts of pre-existing resistant strains
Adverse effect profile	Similar to BP alone	Similar to BP alone; however, there are isolated reports of pseudomembranous colitis associated with topical clindamycin
Stability	Unstable at high temperatures; requires compounding by pharmacist and refrigeration by patient (single-use packaging circumvents this issue)	Stable at room temperature; some products require compounding; does not require refrigeration
Availability	By prescription only	By prescription only

tion by a pharmacist at the time the medication is dispensed, and the patient is advised to discontinue use 2 months after the date of collection. Other formulations do not require reconstitution prior to dispensing. These newer formulations have been shown to exhibit similar efficacy to the older versions, with similar adverse effects.[28]

4. Summary

The use of benzoyl peroxide in combination with topical antibiotics confers specific advantages over topical antibacterials alone in the treatment of acne: (i) combination products exhibit greater *in vivo* activity in decreasing follicular colonization by *P acnes* and lesion counts than either product used alone; (ii) benzoyl peroxide-containing combination products prevent the development of antibiotic resistance; and (iii) combination products containing benzoyl peroxide can bring about significant clinical improvement in patients with acne who have already developed antibacterial resistance. Table I provides a comparison of the available benzoyl peroxide combination therapies.

Benzoyl peroxide-containing combination products are an important part of topical acne therapy and, in light of the potential for scarring and recent issues with emerging antibiotic resistance, should be considered, along with topical retinoids, as cornerstones in the treatment of this most common and potentially physically and psychologically damaging skin disorder.

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